Mehboob Alam

Las Cruces, USA mehboob@nmsu.edu
Linkedin Google Scholar ResearchGate

EDUCATION

MS in Mathematics August 2025 - Present
New Mexico State University, Las Cruces, USA

- ✓ *GPA*: 3.5/4.0
- ✓ *Related coursework*: Advanced Linear Algebra, Abstract Algebra-I, Introduction to Real Analysis-I & II, Probability: Theory and Applications, Statistics: Theory and Applications, Measure and Integration and Machine Learning I

MS in Engineering Sciences (Applied Mathematics) February 2022 - January 2024
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan

- ✓ *GPA*: 3.75/4.0
- ✓ *Related coursework*: Analytical Solution of Partial Differential Equations for Engineers and Computational Methods for Engineers
- ✓ *Thesis:* Radial basis function collocation method for fractional partial differential equations
- ✓ *Advisor:* Prof. Dr. Sirajul Haq

BS in Mathematics September 2016 - November 2020
University of Peshawar, Peshawar, Pakistan

- ✓ *GPA*: 3.72/4.0
- ✓ *Related coursework*: Major in Mathematics with minor courses in Physics, Probability and Statistics
- ✓ *Thesis:* Stability analysis of impulsive fractional and q-fractional implicit differential equations
- ✓ *Advisor:* Prof. Dr. Akbar Zada

RESEARCH FIELDS AND INTERESTS

- ✓ Fractional differential equations
- ✓ Stochastic differential equations
- ✓ Numerical analysis & scientific computing
- ✓ Machine learning
- ✓ Mathematical biology

PUBLICATIONS

Journal Articles

- ✓ Zada, A., Riaz, U., Jamshed, J., Alam, M., & Kallekh, A. (2025). Analysis of impulsive Caputo fractional integro-differential equations with delay. *Mathematical Methods in the Applied Sciences*, **48**(2), 2102-2121.
- ✓ Alam, M., Zada, A., & Abdeljawad, T. (2024). Stability analysis of an implicit fractional integro-differential equation via integral boundary conditions. *Alexandria Engineering Journal*, **87**, 501-514.
- ✓ Alam, M., Haq, S., Ali, I., Ebadi, M. J., & Salahshour, S. (2023). Radial Basis Functions Approximation Method for Time Fractional FitzHugh-Nagumo Equation. *Fractal and Fractional*, **7**(12), 882.

- ✓ Amin, F., Waqas, H. A., Alam, M., Yousafzai, A., Naeem, K., Khan, N., Gerges, M., & Fawad, M. (2023). Strengthening of Reinforced Concrete Deep Beams with Large Openings Using Metal Plates, *Emirati Journal of Civil Engineering and Applications*, **1**(1), 31-44.
- ✓ Alam, M., Zada, A., Begum, S., & Riaz, U. (2023). Analysis of Fractional Integro-differential System with Impulses. *International Journal of Applied and Computational Mathematics*, **9**, 93.
- ✓ Majeed, R., Zhang, B., & Alam, M. (2023). Fractional Langevin Coupled System with Stieltjes Integral Conditions. *Mathematics*, **11**, 2278.
- ✓ Alam, M., & Khalid, K. H. (2023). Analysis of q-fractional coupled implicit systems involving the nonlocal Riemann–Liouville and Erdélyi-Kober q-fractional integral conditions. *Mathematical Methods in the Applied Sciences*, **46**(12), 12711-12734.
- ✓ Alam, M., Khan, A., & Asif, M. (2023). Analysis of implicit system of fractional order via generalized boundary conditions. *Mathematical Methods in the Applied Sciences*, **46**(9), 10554–10571.
- ✓ Guo, L., Riaz, U., Zada, A., & Alam, M. (2023). On Implicit Coupled Hadamard Fractional Differential Equations with Generalized Hadamard Fractional Integro-Differential Boundary Conditions. *Fractal and Fractional*, **7**(1), 13.
- ✓ Zhang, B., Majeed, R., & Alam, M. (2022). On Fractional Langevin Equations with Stieltjes Integral Conditions. *Mathematics*, **10**, 3877.
- ✓ Zada, A., Alam, M., Khalid, K. H., Iqbal, R., & Popa, I. L. (2022). Analysis of Q-Fractional Implicit Differential Equation with Nonlocal Riemann–Liouville and Erdélyi-Kober Q-Fractional Integral Conditions. *Qualitative Theory of Dynamical Systems*, **21**(93), 1-39.
- ✓ Alam, M., & Zada, A. (2022). Implementation of q-calculus on q-integro-differential equation involving anti-periodic boundary conditions with three criteria. *Chaos, Solitons & Fractals*, **154**, 111625.
- ✓ Alam, M., Zada, A., & Riaz, U. (2021). On a coupled impulsive fractional integrodifferential system with Hadamard derivatives. *Qualitative Theory of Dynamical Systems*, **21**(8), 1-31.
- ✓ Alam, M., Zada, A., Popa, I. L., Kheiryan, A., Rezapour, S., & Kaabar, M. K. A. (2021). A Fractional Differential Equation with Multi-Point Strip Boundary Condition involving the Caputo Fractional Derivative and its Hyers–Ulam Stability. *Boundary Value Problems*, **2021**, 73.
- ✓ Alam, M., & Shah, D. (2021). Hyers–Ulam stability of coupled implicit fractional integro-differential equations with Riemann–Liouville derivatives. *Chaos, Solitons & Fractals*, **150**, 111122.
- ✓ Luo, D., Alam, M., Zada, A., Riaz, U., & Luo, Z. (2021). Existence and stability of implicit fractional differential equations with Stieltjes boundary conditions having Hadamard derivatives. *Complexity*, **2021**(3), 1-36.
- ✓ Wang, X., Alam, M., & Zada, A. (2020). On coupled impulsive fractional integro-differential equations with Riemann–Liouville derivatives. *AIMS Mathematics*, **6**(2), 1561-1595.
- ✓ Zada, A., Alam, M., & Riaz, U. (2021). Analysis of q-fractional implicit boundary value problems having Stieltjes integral conditions. *Mathematical Methods in the Applied Sciences*, **44**(6), 4381–4413.

Journal Articles Submitted

- ✓ RBFs approximation method for time fractional Rosenau-Hyman equation.
- ✓ Analysis of nonlocal implicit system of fractional order via Riemann–Liouville integral conditions.

RESEARCH PROJECTS

RBF collocation method for fractional PDEs

- ✓ Applied and studied a meshfree RBF collocation method for solving time-fractional partial differential equations
- ✓ Implemented a finite difference scheme for temporal discretization with an error order of $\mathcal{O}(\Delta t^{2-\alpha})$
- ✓ Achieved high accuracy and efficiency, as demonstrated by numerical experiments and error analysis
- ✓ Published a research article on the application of RBFs to the time-fractional FitzHugh-Nagumo equation
- ✓ Utilized MATLAB for computer simulations and analysis

Finite element analysis of reinforced concrete deep beams

- ✓ Utilized Abaqus software to conduct finite element analysis of reinforced concrete deep beams with large openings
- ✓ Investigated the impact of different metal plate orientations on structural behavior and load-carrying capacity
- ✓ Identified the most effective metal plate configuration (case 3) in terms of deflection and load-carrying capacity
- ✓ Analyzed the cost-effectiveness of different concrete grades to determine the optimal design
- ✓ Developed a practical and secure structural solution for strengthening deep beams
- ✓ Contributed to the understanding of deep beam behavior and design under various loading conditions
- ✓ Identified potential areas for future research, such as dynamic load analysis and alternative plate configurations

A hybrid approach to resource allocation in railway networks

- ✓ Implemented a hybrid approach combining Parabolic Data Envelopment Analysis (PDEA) and Sequential Algorithm (SA) for resource allocation in railway stations
- ✓ Developed an integer-based PDEA frontier to ensure practical implementation of resource allocation
- ✓ Contributed to the understanding of railway efficiency in emerging economies, particularly Pakistan
- ✓ Improved overall system efficiency from 56% to 99.5% by reallocating staff among 44 railway stations in KPK
- ✓ Utilized DEAFrontier software to conduct the analysis

Qualitative analysis of fractional and q-fractional differential equations

- ✓ Implemented fixed point theorems to establish existence and uniqueness of solutions for impulsive fractional and q-fractional differential equations
- ✓ Utilized stability analysis techniques (e.g., Hyers-Ulam, Hyers-Ulam-Rassias) to assess the stability of solutions
- ✓ Developed conditions for the existence and uniqueness of solutions under various assumptions
- ✓ Established stability criteria for the considered equations
- ✓ Demonstrated the effectiveness of the proposed methods through examples and simulations
- ✓ Utilized MATLAB and Mathematica for computer simulations and analysis

PROFESSIONAL QUALIFICATION

Primary Teaching Certificate 2016

Allama Iqbal Open University, Islamabad, Pakistan

- ✓ *Marks:* 73%

- ✓ *Key Training:* Received training on constructivist teaching methods, effective classroom management, assessment strategies, technology integration in education and curriculum development

EXPERIENCE

Teaching Assistant August 2025 - Present
New Mexico State University, Las Cruces, USA
- ✓ *MATH 3460:* Introduction to ODEs — Spring 2026, Fall 2026
- ✓ *MATH 2415:* Introduction to Linear Algebra — Fall 2025

Lecturer February 2024 – March 2025
The University of Lahore, Sargodha, Pakistan
- ✓ *MATH-402:* Partial Differential Equations — Spring 2024
- ✓ *MAT07301:* Complex Analysis — Spring 2024
- ✓ *MAT12145:* Introduction to Mathematical Modeling — Spring 2024, Fall 2024
- ✓ *MAT09212/CS09202:* Discrete Mathematics/Structure — Spring 2024, Fall 2024
- ✓ *MATH2100/MA2334:* Ordinary Differential Equations — Summer 2024
- ✓ *MATH-204:* Metric Space and Group Theory — Summer 2024
- ✓ Infinite Series and Sequence — Summer 2024
- ✓ Probability and Statistics — Summer 2024
- ✓ Multivariate Calculus — Fall 2024
- ✓ *Other responsibilities:* Served as timetable coordinator, assisting with academic scheduling and course coordination.

Teaching Assistant February 2022 - January 2024
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan
- ✓ *ES202:* Engineering Statistics — Spring 2022
- ✓ *MT203:* Complex Variables and Transforms — Spring 2022, Spring 2023
- ✓ *MT102:* Differential Equations and Linear Algebra I — Summer 2022
- ✓ *MT101:* Calculus I — Fall 2022
- ✓ *ES341:* Numerical Analysis — Fall 2022, Fall 2023

Visiting Lecturer November 2021 - February 2022
Government Postgraduate College Nowshera, Nowshera, Pakistan
- ✓ Basic Algebra — Fall 2021
- ✓ Differential Geometry — Fall 2021
- ✓ Partial Differential Equations — Fall 2021

Visiting Lecturer March 2021 - August 2021
Government Degree College Pabbi Nowshera, Nowshera, Pakistan
- ✓ Basic Algebra — Spring 2021
- ✓ Partial Differential Equations — Spring 2021
- ✓ Measure Theory — Spring 2021
- ✓ Classical Mechanics — Spring 2021

PRESENTATIONS

- ✓ Existence and Uniqueness of Differential Equation in the course of Analytical Solution of Partial Differential Equations for Engineers, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan (17 December 2022)
- ✓ A simple non-chaotic map generating subdiffusive, diffusive, and superdiffusive dynamics in the course of Special Topics in Applied Mathematics, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan (06 December 2022)

- ✓ Analysis of Parabolic DEA and Sequential Algorithm: A Case Study in KPK Railway Stations in the course of Operation Management, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan (02 June 2022)
- ✓ Understanding the performance measurement of Indian steel Industry: A DEA approach in the course of Operation Management, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan (06 April 2022)
- ✓ Improving inland freight logistic efficiencies: Is there any ideal modal split? in the course of Operation Management, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan (09 March 2022)

CONFERENCES

- ✓ Attended the 13th Conference on Dynamical Systems Applied to Biology and Natural Sciences, organized by the Mathematical and Theoretical Biology Group at the Basque Center for Applied Mathematics, Bilbao, Spain (8-11 February 2022)
- ✓ Attended the International Conference on Recent Trends in Applied Mathematics, organized by the Department of Mathematics, University of Engineering and Technology, Lahore, Pakistan (18 May 2022)

AWARDS & DISTINCTION

- ✓ **Certificate of Merit** 2020
  University of Peshawar, Peshawar, Pakistan
- ✓ **Laptop by Higher Education Commission of Pakistan** 2019
  University of Peshawar, Peshawar, Pakistan

SCHOLARSHIPS/FUNDING

- ✓ **Graduate Assistantship Scholarship** Fall 2026 - Spring 2027
  New Mexico State University, Las Cruces, USA
- ✓ **Graduate Assistantship Scholarship** Fall 2025 - Spring 2026
  New Mexico State University, Las Cruces, USA
- ✓ **Graduate Assistantship Scholarship** 2022 - 2024
  GIK Institute of Engineering Sciences and Technology, Topi, Pakistan
- ✓ **Ehsaas Scholarship** 2019 - 2020
  University of Peshawar, Peshawar, Pakistan

SKILLS

Computational

- ✓ MATLAB, Python, Julia, Mathematica, Abaqus and DEAFrontier

Typesetting

- ✓ Microsoft Office and LaTeX

Other Skills

- ✓ Enjoy all sports particularly cricket, football and badminton. Love to travel and experience different cultures

VOLUNTEER & LEADERSHIP ACTIVITIES

Finance Secretary August 2025 – Present
Pakistani Students Association (PSA), NMSU

- ✓ Managed budget planning and expense tracking for student-led events and activities

- ✓ Coordinated with executive committee to ensure transparent financial reporting
- ✓ Supported cultural, academic, and community-building initiatives for Pakistani students

Learn-A-Palooza November 2025
New Mexico State University, Las Cruces, USA

- ✓ Volunteered with NMSU STEM Outreach Center for campus-wide educational event
- ✓ Assisted in event coordination, student engagement, and logistics
- ✓ Completed 3 hours of certified community service

Volunteer Judge – Data Science & Analytics Competition February 2026
New Mexico State University, Las Cruces, USA

- ✓ Evaluated student projects in data science and analytics during the NM TSA State Conference
- ✓ Assessed analytical approaches and presentation quality and provided feedback to participants